

MHSDS integration into pipeline

DS2_Task4

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1. Introduction

- This report documents the design, development and implementation of an automated data pipeline for the Mental Health Services Data Set (MHSDS) published monthly by NHS England.
- MHSDS is an official national dataset that captures information about people receiving NHS-funded secondary mental health, learning disability and autism services across England.
- The objective of this task was to add MHSDS monthly data as a new pipeline source and automate the monthly append process so that new data is ingested with minimal manual effort.
- This pipeline was developed as part of a group data science internship project, representing the fourth task in the series following hypothesis testing, clustering segmentation and anomaly detection.

2. Objective

- To identify and access the correct MHSDS monthly data files published by NHS England.
- To build a Python-based pipeline that downloads, extracts, cleans and standardises the monthly CSV data.
- To implement a deduplication mechanism that prevents the same month's data from being loaded more than once.
- To append each new month's data to a growing master dataset automatically.
- To document a clear repeatable process for monthly pipeline execution going forward.

3. Data Source

- **Publisher:** NHS England (formerly NHS Digital)
- **Dataset:** Mental Health Services Data Set (MHSDS)
- **Publication frequency:** Monthly
- **Geographic coverage:** England

- **Access method:** Manual URL retrieval from the NHS publication page followed by automated ZIP download
- **Publication page:** [Mental Health Services Monthly Statistics](#)
- **File format:** ZIP archive containing CSV file
- **Data loaded:** Performance February 2026 (latest available at time of development)

4. Data Description

The MHSDS monthly data file contains the following fields:

- REPORTING_PERIOD_START - First day of the reporting month
- REPORTING_PERIOD_END - Last day of the reporting month
- STATUS - Type of submission (e.g. Performance)
- BREAKDOWN - Geographic or demographic breakdown level (e.g. Provider, Sub ICB)
- PRIMARY_LEVEL - Organisation code
- PRIMARY_LEVEL_DESCRIPTION - Full organisation name
- SECONDARY_LEVEL - Sub-breakdown code
- SECONDARY_LEVEL_DESCRIPTION - Sub-breakdown name
- MEASURE_ID - Unique metric code (e.g. MHS01, MH10a)
- MEASURE_NAME Full description of the metric being measured
- MEASURE_VALUE - The recorded value for that metric (may be suppressed)

Key statistics of the February 2026 file:

- Total rows: 771,626
- Total unique measures: 378
- Total unique NHS organisations: 1,149
- Suppressed values (marked as *): 619,013 rows
- File size after processing: 202.7 MB

5. Pipeline Architecture

The pipeline was built entirely in Python within a notebook and consists of the following stages:

- **Stage 1 - URL Construction:** Builds the NHS publication page URL dynamically using the target month and year following the pattern performance-{month}-{year}

```
BASE_URL = (
    "https://digital.nhs.uk/data-and-information/publications/statistical/"
    "mental-health-services-monthly-statistics/performance-{month}-{year}"
)
```

- **Stage 2 - URL Retrieval:** Looks up the direct ZIP download URL from a manually maintained KNOWN_URLS dictionary, bypassing NHS website scraping restrictions (403 Forbidden)

```
*** Found URL for 2026-02
ZIP URL: https://files.digital.nhs.uk/CD/549410/MHSDS%20Data_FebPerf_2026.zip
```

- **Stage 3 - Download & Extraction:** Downloads the ZIP file using the requests library and extracts the largest CSV from inside the archive using the zip file module entirely in memory without saving to disk

```
*** Downloading ZIP from: https://files.digital.nhs.uk/CD/549410/MHSDS%20Data_FebPerf_2026.zip
Downloaded 4.1 MB
Extracting: MHSDS Data_FebPerf_2026/MHSDS Data_FebPerf_2026.csv
Loaded 771,626 rows x 11 columns
Shape: (771626, 11)
Index(['REPORTING_PERIOD_START', 'REPORTING_PERIOD_END', 'STATUS', 'BREAKDOWN',
       'PRIMARY_LEVEL', 'PRIMARY_LEVEL_DESCRIPTION', 'SECONDARY_LEVEL',
       'SECONDARY_LEVEL_DESCRIPTION', 'MEASURE_ID', 'MEASURE_NAME',
       'MEASURE_VALUE'],
      dtype='object')
```

- **Stage 4 - Cleaning & Standardisation:** Strips whitespace from columns and values, parses date columns from text to datetime format, splits the MEASURE_VALUE column into three derived columns (MEASURE_VALUE_RAW, MEASURE_VALUE_NUMERIC, SUPPRESSED) and adds a PIPELINE_INGESTED_AT timestamp

Cleaned: 771,626 rows
Suppressed rows: 619,013

```
Index(['REPORTING_PERIOD_START', 'REPORTING_PERIOD_END', 'STATUS', 'BREAKDOWN',
       'PRIMARY_LEVEL', 'PRIMARY_LEVEL_DESCRIPTION', 'SECONDARY_LEVEL',
       'SECONDARY_LEVEL_DESCRIPTION', 'MEASURE_ID', 'MEASURE_NAME',
       'MEASURE_VALUE_RAW', 'MEASURE_VALUE_NUMERIC', 'SUPPRESSED',
       'PIPELINE_INGESTED_AT'],
      dtype='object')
```

	REPORTING_PERIOD_START	REPORTING_PERIOD_END	STATUS	BREAKDOWN	PRIMARY_LEVEL	PRIMARY_LEVEL_DESCRIPTION	SECONDARY_LEVEL	SECONDARY_LEVEL_DESCRIPTION	MEASURE_ID	MEASURE_NAME	MEASURE_VALUE_RAW	MEASURE_VALUE_NUMERIC	SUPPRESSED	PIPELINE_INGESTED_AT
0	2026-02-01	2026-02-28	Performance	Sub ICB - GP Practice or Residence	93C	NHS NORTH CENTRAL LONDON ICB - 93C	NONE	NONE	MH10a	People subject to CTO or conditional discharge	*	NaN	True	2026-05-15 13:19:21
1	2026-02-01	2026-02-28	Performance	Sub ICB - GP Practice or Residence	93C	NHS NORTH CENTRAL LONDON ICB - 93C	NONE	NONE	MH11	People subject to a short term order (mental h...	5	5.0	False	2026-05-15 13:19:21
2	2026-02-01	2026-02-28	Performance	Sub ICB - GP Practice or Residence	93C	NHS NORTH CENTRAL LONDON ICB - 93C	NONE	NONE	MH101	People in contact with services at the end of ...	64440	64440.0	False	2026-05-15 13:19:21

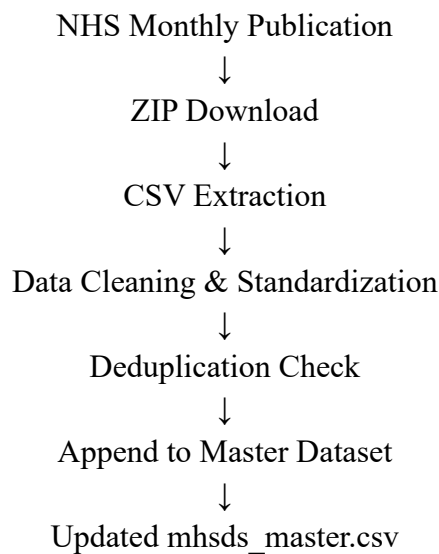
- **Stage 5 - Deduplication Check:** Before appending, checks whether the current source file key (e.g. performance-2026-02) already exists in the master dataset - skips if already loaded

```
*** Source key:performance-2026-02
Master file does not exist yet this will be the first load
Already loaded? : False
```

- **Stage 6 - Master Append:** Appends the cleaned data to mhsds_master.csv with headers written only on first load. Tags every row with a SOURCE_FILE column for traceability

```
*** Master file does not exist yet this will be the first load
Appended 771,626 rows → master file now 202.7 MB
'performance-2026-02' appended to master successfully!
Master file: mhsds_master.csv (202.7 MB)
```

Pipeline Architecture



6. Key Design Decisions

- **Source-key based deduplication instead of date-based:** During development it was discovered that each monthly file contains data from multiple reporting periods including revised historical months. Using the source file key (performance-YYYY-MM) rather than the reporting date as the deduplication key correctly handles this without creating duplicate rows.
- **In-memory ZIP extraction:** The ZIP file is downloaded and extracted entirely in memory using io.BytesIO no intermediate files are written to disk, keeping the pipeline clean and efficient.
- **Suppressed value handling:** NHS England suppresses small patient counts (shown as *) to protect patient privacy. The pipeline preserves the original value in MEASURE_VALUE_RAW, converts numeric values to MEASURE_VALUE_NUMERIC (NaN where suppressed) and adds a boolean SUPPRESSED flag giving analysts full flexibility.
- **Manual URL store (KNOWN_URLS):** The NHS website returned a 403 Forbidden error blocking automated scraping. The solution adopted was a manually maintained

dictionary of direct download URLs a reliable and transparent approach appropriate for this project scope.

- **SOURCE_FILE column:** Every row in the master dataset is tagged with which monthly file it came from, enabling easy filtering by publication month.

7. Tools & Technologies

- Language: Python 3
- Environment: Google Colab (Jupyter Notebook)
- Libraries used: pandas - data loading, cleaning and manipulation
- requests - HTTP download of ZIP files
- zipfile - in-memory ZIP extraction
- io - BytesIO buffer for in-memory file handling
- python-dateutil - month offset calculations
- datetime - timestamps and date handling
- os - file existence checks and path operations
- Storage: Google Drive
- Data Format: ZIP archive containing CSV

8. Results

- Pipeline successfully built and tested end to end.
- February 2026 MHSDS data loaded into master dataset:
 - 771,626 rows ingested
 - 378 unique mental health measures captured
 - 1,149 NHS organisations represented
 - Master file size: 202.7 MB
- Deduplication confirmed working re-running the pipeline for February 2026 correctly skips without creating duplicate rows.
- run_latest() function successfully auto-detects the latest available month and runs the full pipeline with a single function call.
- Master dataset (mhsds_master.csv) saved permanently to Google Drive.

```
*** Found URL for 2026-02
Downloading ZIP from: https://files.digital.nhs.uk/CD/549410/MHSDS%20Data\_FebPerf\_2026.zip
Downloaded 4.1 MB
Extracting: MHSDS Data_FebPerf_2026/MHSDS Data_FebPerf_2026.csv
Loaded 771,626 rows x 11 columns
Cleaned: 771,626 rows
'performance-2026-02' already exists in master will skip
February 2026 already loaded (skipping)

Pipeline returned: False
```

Pipeline validated with live NHS data - On 16 May 2026

- when NHS England published March 2026 MHSDS data,
- the pipeline was tested by adding the March URL to KNOWN_URLS
- and running run_latest(). March 2026 data (764,377 rows)
- was automatically downloaded, cleaned and appended to the
- master dataset without any code changes growing the
- master file from 202.7 MB (771,626 rows) to 403.1 MB
- (1,536,003 rows), confirming the monthly automation works
- correctly in production with real live NHS data.

Note :-The March 2026 append was performed as a validation

- test only. The submitted mhsds_master.csv contains
- February 2026 data only (771,626 rows, 202.7 MB) as the
- primary pipeline deliverable. March 2026 data can be
- appended at any time by adding its URL to KNOWN_URLS
- in Cell 5 and running run_latest().

9. Monthly Automation Process

The following process should be followed each month when NHS England publishes new MHSDS data:

- Visit the NHS Mental Health Services Monthly Statistics publication page
- Click the latest month's publication
- Right-click the "MHSDS Data File" download button and copy the link address
- Paste the URL into the KNOWN_URLS dictionary in the notebook
- Run run_latest() Function the pipeline will automatically download, extract, clean, deduplicate and append the new month's data to the master dataset.

```
*** Today's date: 2026-05-15
Checking for latest available MHSDS data...

Trying April 2026...
No URL in KNOWN_URLS for 2026-04 skipping.
Add the download URL to KNOWN_URLS to load this month.

Trying March 2026...
No URL in KNOWN_URLS for 2026-03 skipping.
Add the download URL to KNOWN_URLS to load this month.

Trying February 2026...
Found URL for 2026-02: https://files.digital.nhs.uk/CD/549410/MHSDS%20Data_FebPerf_2026.zip
Downloading ZIP from: https://files.digital.nhs.uk/CD/549410/MHSDS%20Data_FebPerf_2026.zip
Downloaded 4.1 MB
Extracting: MHSDS Data FebPerf_2026/MHSDS Data_FebPerf_2026.csv
Loaded 771,626 rows x 11 columns
Cleaned: 771,626 rows
'performance-2026-02' already exists in master will skip
February 2026 already loaded (skipping)

Summary:
No new data was appended.
Either all recent months are already loaded,
or their URLs are missing from KNOWN_URLS
```

```

Total rows:771,626
Total columns: 16
File size: 280.7 MB
Months loaded: 1
Unique measures:178
Unique orgs: 1149
Suppressed rows:435,413
Date range:2025-01-01 - 2026-02-01

Loaded source files:
SOURCE_FILE
performance-2026-02 771626

Breakdown types available:
BREAKDOWN
Provider: 151001
Provider: Ethnicity 101394
Provider: Age Band 53122
Provider: Upper Ethnicity 42672
Provider: IMD Decile 32494
Provider: ID Status 26586
Provider: Autism Status 26586
Provider: ContactChannel:GP 25539
Sub ICB - GP Practice or Residence 20972
Provider: Gender 17724

Sample measures (first 10):
MEASURE_ID MEASURE_NAME
0 MH10a People subject to CTO or conditional discharge (mental health services) at the end of the reporting period, aged 0 to 17
1 MH11 People subject to a short term order (mental health services) at the end of the reporting period
2 MH01 People in contact with services at the end of the reporting period
3 MH07 People with an open hospital spell at the end of the reporting period
4 MH07a People with an open hospital spell at the end of the reporting period aged 0 to 16
5 MH07b People with an open hospital spell at the end of the reporting period aged 16 to 64
6 MH07c People with an open hospital spell at the end of the reporting period aged 65 and over
7 MH08 People subject to the Mental Health Act at the end of the reporting period
8 MH08a People subject to the Mental Health Act at the end of the reporting period, aged 0 to 17
9 MH09 People subject to detention at the end of the reporting period

first 3 rows:
REPORTING_PERIOD_START REPORTING_PERIOD_END STATUS BREAKDOWN PRIMARY_LEVEL PRIMARY_LEVEL_DESCRIPTION SECONDARY_LEVEL SECONDARY_LEVEL_DESCRIPTION MEASURE_ID MEASURE_NAME MEASURE_VALUE_RAW MEASURE_VALUE_NUMERIC SUPPRESSED PIPELINE_INGESTED_AT SOURCE_FILE
0 2026-02-01 2026-02-28 Performance Sub ICB - GP Practice or Residence 93C NHS NORTH CENTRAL LONDON ICB - 93C NONE NONE MH10a People subject to CTO or conditional discharge. 0 NaN True 2026-05-15 13:19:21 performance-2026-02
1 2026-02-01 2026-02-28 Performance Sub ICB - GP Practice or Residence 93C NHS NORTH CENTRAL LONDON ICB - 93C NONE NONE MH11 People subject to a short term order (mental h. 5 5.0 False 2026-05-15 13:19:21 performance-2026-02
2 2026-02-01 2026-02-28 Performance Sub ICB - GP Practice or Residence 93C NHS NORTH CENTRAL LONDON ICB - 93C NONE NONE MH01 People in contact with services at the end of 64440 64440.0 False 2026-05-15 13:19:21 performance-2026-02

```

10. Deliverables

The following deliverables were successfully produced as part of the MHSDS pipeline integration task:

[mhsds_pipeline.ipynb](#)

Complete notebook-based pipeline module containing:

- automated monthly data ingestion
- ZIP extraction
- preprocessing and standardization
- suppressed value handling
- deduplication logic
- monthly append automation
- master dataset update workflow

[mhsds_master.csv](#)

Final centralized master dataset generated by the pipeline containing appended MHSDS monthly records from NHS England publications.

- Automated monthly append workflow
- The pipeline was designed to support future monthly NHS MHSDS releases by allowing new publication URLs to be added and processed automatically through a single function call.
- February 2026 processed dataset integration
- Successfully integrated the February 2026 MHSDS publication into the historical master dataset without generating duplicate records.

11. Limitations

- **Manual URL entry required each month:**

Due to the NHS website returning a 403 Forbidden error for automated requests, the direct download URL for each new monthly publication must be manually retrieved from the NHS website and added to the KNOWN_URLS dictionary. A fully automated end-to-end pipeline would require either NHS API access or a solution to bypass the access restriction.

- **Suppressed values limit analysis:**

Over 619,013 rows (approximately 80% of the dataset) contain suppressed values marked as *. These cannot be used in numerical analysis, which significantly limits the scope of any downstream statistical or machine learning work on this dataset.

- **KNOWN_URLS must be maintained manually:**

If a month's URL is not added to the dictionary, the pipeline will skip it silently. There is currently no alerting mechanism to notify the user when a new publication is available on the NHS website.

- **Single month per run:**

The pipeline processes one month at a time. If multiple months are missed (e.g. due to inactivity), each month's URL must be added individually and the pipeline run separately for each.

12. Conclusion

- The MHSDS pipeline module was successfully designed, built and tested within the allocated timeframe.
- The pipeline correctly fetches, extracts, cleans and appends monthly MHSDS data to a growing master dataset with full deduplication and audit traceability.
- The architecture is simple, transparent and maintainable requiring only a single URL addition and one function call to ingest each new month of data.
- The master dataset is now ready to serve as a reliable data source for future analytical tasks including trend analysis, performance monitoring and population health reporting across NHS mental health services in England.